

Software System Design (2025) - Architecture

Assignment 2

(40 Points)

You are responsible to design a software architecture for the following greenfield system for an emerging domain. You should use the Attribute Driven Design (ADD) method and document the final architecture in multiple views. In this assignment, FOUR students need to form a group to collectively complete the assignment. The score of each student on this assignment will be determined with the group performance by considering individual contribution to plus reflection on the work.

DEADLINE: 12:00am on (Monday) June 30, 2025

The Digital Twin Platform Case Study provided as the appendix reading for this assignment presents a digital twin platform—a computational platform with digital representations of assets and processes.

Assignment Deliverables

A single PDF file should be submitted per group. Your report should be in TWENTY pages (A4 size) or shorter and include the following deliverables:

- A report on how you have followed the Attribute Driven Design method (24 points). You should perform at least **THREE** iterations (8 points per iteration) when designing the architecture. This part includes two components:
 - A list of inputs for the digital twin platform as the architectural drivers (Step 1), including the Design purpose, Primary functional requirements, Quality attribute scenarios with priority, Constraints, and Architectural concerns (if existing).
 - A list of all the goals of each Iteration (Step 2), as well as the results of Step 3-7 when using the ADD method. This should include all intermediary candidates that might not have been included in the final architecture. A *brief* explanation of the choice should be provided.
- Final software architecture documentation (10 points):
 - Views (models) and cross-views following the template of Views & Beyond (Chapter 18 in 3rd Edition or Chapter 22 in 4th Edition) [1]. There should be at least **THREE** views (with at least one module

view) plus one cross-view. The “**rationale**” part cannot be omitted. (For more information, see [2])

- You are encouraged to use suitable UML diagrams. (For more information, see [3]) However, you may instead choose to use other alternative modeling language notations that you feel more comfortable with.
- Individual remark should be completed by each student (about half page each) in a group (5 points), and contains two components:
 - A description of your personal experiences using the ADD method.
 - A summary of your own contributions to the group work and report.

Notes:

1. You can compile and submit your report in either Chinese or English. You are not necessarily 100% correct in language when using English. However, you should make sure you convey everything in a clear and understandable manner.

2. When you require more information on requirements or affecting factors to make a design decision, you could make a reasonable assumption, document it and then make your design decision.

3. There is no minimum length requirement for this assignment, but it should not exceed TWENTY pages. Be concise and to the point.

4. You will be assessed based on your ability to apply the design method and document the design rather than the absolute correctness of your design.

[1] Software Architecture in Practice, 3rd or 4th Edition (reprint from the English edition published by Tsinghua University Press or China Machine Press)

[2] Documenting Software Architectures: Views and Beyond, 2nd Edition (some e-book available on Internet)

[3] UML Distilled: A Brief Guide to the Standard Object Modeling Language, 2nd Edition (some e-book available on Internet)

Appendix

Digital Twin Platform

(a case study)

Note: this is a greenfield system for an emerging domain in an Industry 4.0 context.

Business Case

The client in this case study is a large food producer with dozens of production plants, many of them containing several production lines. The company is committed to developing innovative solutions as part of its mission to ensure the advancement of food security and promotion of sustainability. For this company, creating a smart factory and automating industrial processes is crucial in addressing the shortage of skilled workers and maintaining affordable prices for consumers.

One of the key investments of the company is creation of a digital twin platform—a computational platform with digital representations of assets and processes. This platform is expected to enable a range of use cases, from simple ones like remote monitoring of industrial processes across plants, to more complex ones such as process simulation and autonomous decision-making to adapt to a changing business environment as quickly as possible. The platform should provide the capabilities to implement various use cases that contribute to the goal of creating a smart factory.

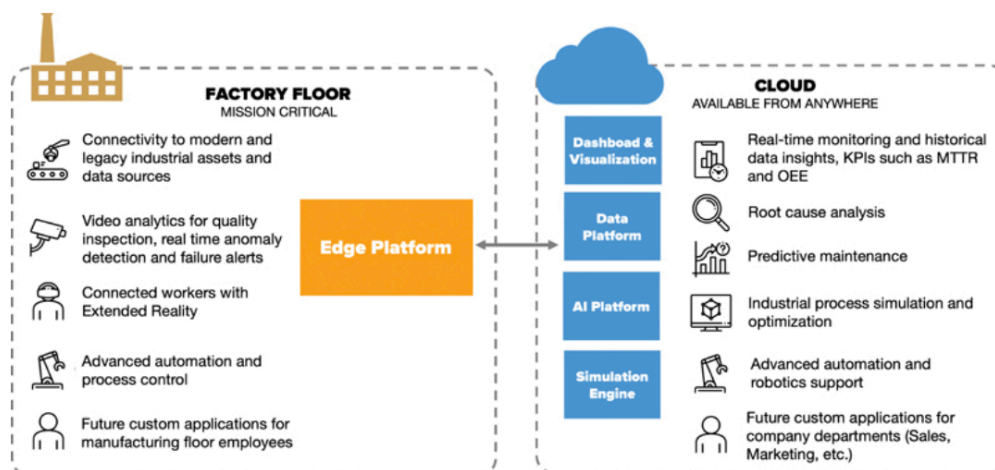


FIGURE 1 Marketecture diagram for the digital twin platform

The marketecture diagram (an informal depiction of the system's structure) shown in the Figure 1 represents the desired solution from a functional perspective.

System Requirements

For requirements elicitation activities, the Digital Twin Maturity Model (DTMM), shown in Figure 2, was used to identify use cases corresponding to each of the five digital twin maturity stages.

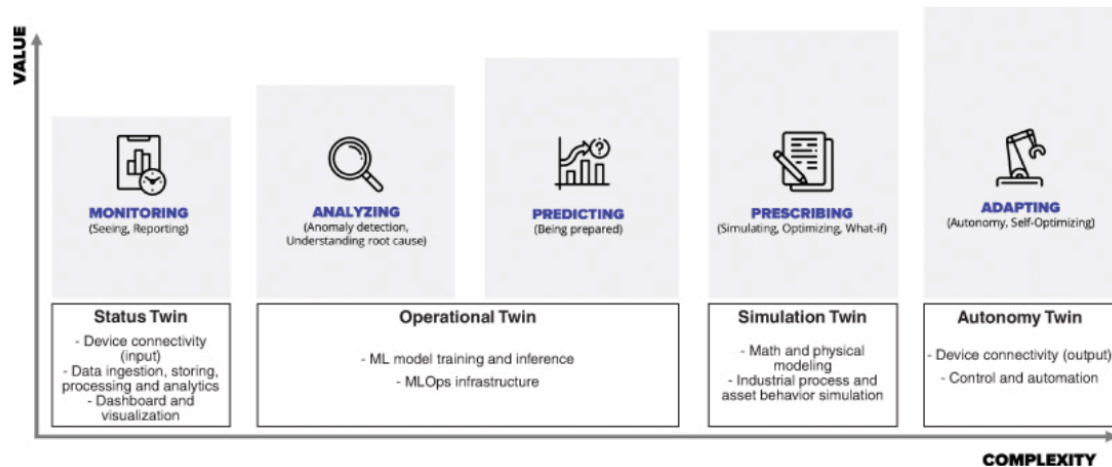


FIGURE 2 Digital Twin Maturity Model

Each stage in this model represents a group of capabilities that the platform should support to provide value to its users. During the requirements elicitation process, base use cases were chosen for each stage, as this technique allows for a holistic view of the system's capabilities rather than focusing on just a part of it. Some of the use cases chosen are abstract and require further development and elicitation to create more concrete forms, as innovative solutions typically require exploration and experimentation. Therefore, the platform's goal is to enable both the identified use cases and similar future ones.

The following subsections summarize the most important requirements collected. These are a set of primary use cases, a set of quality attribute scenarios, and a set of constraints.

1) Use Case Model

TABLE 1 Primary Use Cases for the Digital Twin Platform

Use Case	Description	DTMM Stage
UC-1: Real-time monitoring and historical data insights	As a Global Operator, I can see information such as overall equipment effectiveness (OEE) for any industrial process on any plant. The data should be collected from equipment in real time and stored in the cloud.	Monitoring
UC-2: Quality inspection	As a Quality Control Operator, I can control the quality of production via sensors (including optical sensors). The	Analyzing

automation	system should immediately generate alerts in case of detection of anomalies.	
UC-3: Predictive maintenance and XR for maintenance activities	As a Maintenance Engineer, I can get predictions for equipment failures and diagnose them by accessing historical and real-time information. The information should be accessible through a hands-free extended reality device.	Predicting
UC-4: Process simulation	As a Process Engineer, I can find optimal parameters for equipment setting based on raw materials and desired quality of output. The parameters should be derived from simulation.	Prescribing
UC-5: Advanced automation	As an Automation Engineer, I want to automate operational decision-making, so that the plant can adapt to changing conditions (such as orders, raw materials, or power supply) without human intervention.	Adapting

2) Quality Attribute Scenarios

TABLE 2 Quality Attribute Scenarios for the Digital Twin Platform

ID	Quality Attribute	Scenario	Associated Use Case
QA-1	Availability	Critical system functions for the production floor must remain 100% available in the event of a cloud or network outage. Should such an event occur, data should be cached at the edge for at least 8 hours.	UC-2
QA-2	Security	An unauthorized access attempt is made. The attempt is denied and logged, and the system administrator is alerted within 15 minutes of the attempted breach.	All use cases
QA-3	Deployability	A system deployment is initiated. The deployment is fully automated and supports at least development, test, and production environments.	All use cases
QA-4	Performance	100,000 data points (where a data point is 120 bytes in size, on average) arrive at the system per second. The system processes them all within 1 second.	All use cases
QA-5	Performance	The real-time dashboard for Global Operators is refreshed automatically, at 15-second intervals, with less than 15 seconds data latency.	UC-1
QA-6	Performance	Industrial processes, coupled with real-time anomaly detection, are continuously visually inspected. Inspection results are received within 5 seconds.	UC-2
QA-7	Performance	When a data analyst requests ad hoc SQL-like queries for raw and aggregated historical data (multirelational data that can include various equipment from different plants), 95% of queries return results within 30 seconds.	UC-3, 4, 5
QA-8	Extensibility	New devices and sensors can be added to the system at runtime, with no interruption of ongoing data collection and system functionality.	All use cases

QA-9	Extensibility	New applications (e.g., Production, Quality, Supply Chain, Sales) can be created by building on existing services without the need to re-architect or change the system's main components (such as Edge, Data, and AI Platforms).	All use cases
QA-10	Observability	Key performance metrics and error logs are collected, aggregated, and visualized in a monitoring dashboard with a maximum delay of 5 seconds. All use cases.	All use cases

3) Constraints

TABLE 3 Constraints on the Digital Twin Platform

ID	Constraint
CON- 1	The system shall use existing company cloud infrastructure on AWS.
CON- 2	Use a cloud-native strategy (prefer microservices, containers, and managed services) for building scalable applications.
CON- 3	When selecting cloud services, give preference to the services that have the lowest and most predictable costs.
CON- 4	NIST's IoT cybersecurity capabilities must be implemented to manage cybersecurity risks.